Bluetooth LE Coded PHY White Paper

Revision History

Date	Version	Description
August 6, 2019	0.50	Preliminary version created.
December 2, 2020	0.51	Corrected typos

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This white paper describes the LE Coded PHY aspect of the Bluetooth Low Energy 5 specification and its advantages. It also lists the features of Coded PHY that are supported in Atmosic's ATMxxxx series system-on-a-chip (SoC) solutions, how to enable those features as well as point to a few common use-cases of LE Coded PHY such as beacons. The FAQs section answers a few specific questions that the user might find useful.

Overview

Prior to Bluetooth 5.0, Bluetooth Low Energy supported only 1 Mbps rate known as LE 1M PHY. With Bluetooth 5.0, 2 optional PHYs were introduced. LE 2M with 2 Mbps rate and LE Coded PHY with 500 kbps and 125 kbps rates.

LE Coded PHY allows the user to get longer reach without increasing the transmitter output power. Hence, LE Coded PHY is sometimes also referred to as Bluetooth LE Long Range. The energy per bit is increased compared to uncoded PHY to add redundancy in the transmitted packet. Thus, through the combination of reduced data rate and forward-error-correction (FEC) code, receiver sensitivity is improved to achieve longer range.

The symbol rate for LE Coded PHY is 1 MS/s similar to LE 1M. However, for 125 kbps, each bit is represented by 8 symbols (S=8) and for 500 kbps each bit is represented by 2 symbols (S=2).

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FEC and Pattern Mapping

Both 125 kbps and 500 kbps use ½ rate convolutional encoder with constrained length of 4 to achieve ~3 dB improvement in receiver sensitivity. With 125 kbps, additional improvement in receiver sensitivity is achieved using a 1:4 pattern mapping, where each bit out of the convolutional encoder is mapped to 4 symbols: bit 1 is mapped to symbols 1100 and bit 0 is mapped to 0011. The improvement in sensitivity translates to ~2x increase in range (for line-of-sight) for 125 kbps with respect to 1 Mbps.

Packet Format

The packet format for LE Coded PHY is as shown below (from Figure 2.3 Vol 6, Part B, Section 2.2 of the Bluetooth Core specification), where the preamble and FEC Block 1 (Access Address, 2-bit Coding indicator and 3 termination bits for the convolutional encoder) are encoded at 125 kbps (S=8). Preamble is 80 µs with the symbol pattern 00111100 repeated 10 times. The 2-bit Coding Indicator field tells the receiver whether the payload that follows in FEC Block 2 is encoded at 500 kbps or 125 kbps. The notable difference is that preamble is 80 µs for LE Coded PHY compared to 8 µs in LE 1M.

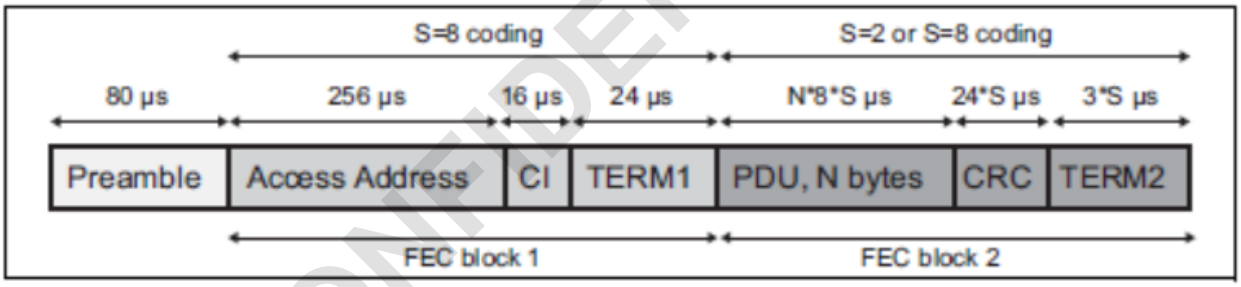


Figure 1 - LE Coded PHY Packet Format as in Bluetooth Core Specification

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Note that only certain types of Protocol Data Units (PDUs) can be transmitted using LE Coded PHY:

- ADV_EXT_IND
- AUX_ADV_IND
- AUX_SYNC_IND
- AUX_CHAIN_IND
- AUX_SCAN_REQ
- AUX_SCAN_RSP
- AUX_CONNECT_REQ
- AUX_CONNECT_RSP

Please refer to Vol 6, Part B, Section 2.3 of the Bluetooth Core 5.1 specification for more details.

PHY Specifications

As per the Bluetooth LE Core spec, the RX sensitivity requirement @BER=0.1% is -75 dBm for 500 kbps and -82 dBm for 125 k bps, compared to -70 dBm for 1 Mbps. All the interference specs for 500 kbps are 4 dB worse with respect to LE 1M and 9 dB worse for 125 kbps. Please refer to Vol 6, Part B, Sections 4.1 and 4.2 of the Bluetooth Core 5.1 specification. Please refer to the datasheet under ../Customer Gateway/Docs_CustomerName_only for the performance of Atmosic's ATMxxx series SOCs that greatly exceeds the Core spec requirements.

Atmosic's ATMxxxx series SoCs support LE Coded PHY (in addition to LE 1M and LE 2M). The improved RX sensitivity of LE Coded PHY with respect to LE 1M, allows the user to achieve a longer range with the same transmit power or reduce the transmit power for the same range, at the cost of reduced throughput due to longer packets for the same amount of data. The following picture shows the Tx power profile for a 37-byte packet transmitted using 4 Bluetooth LE rates. Note that the packet duration for 500 kbps is more than 2 times that of 1 Mbps because the preamble and FEC Block 1 are always transmitted at 125 kbps even for a 500 kbps packet.

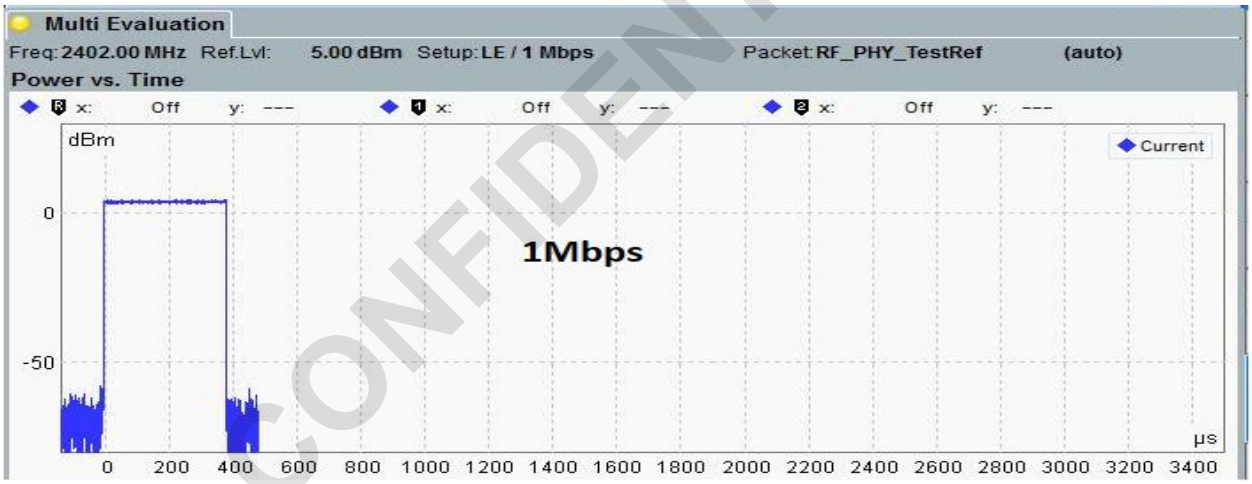
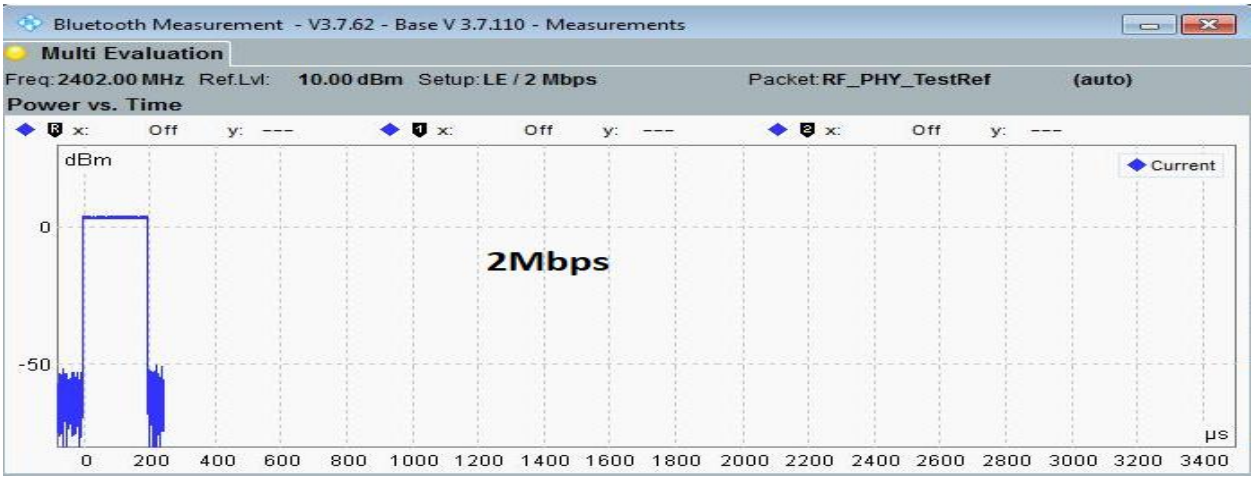


Figure 2 - Tx Power Profile for a 37-Byte Packet Transmitted at 1 Mbps



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Figure 3 - Tx Power Profile for a 37-Byte Packet Transmitted at 2 Mbps

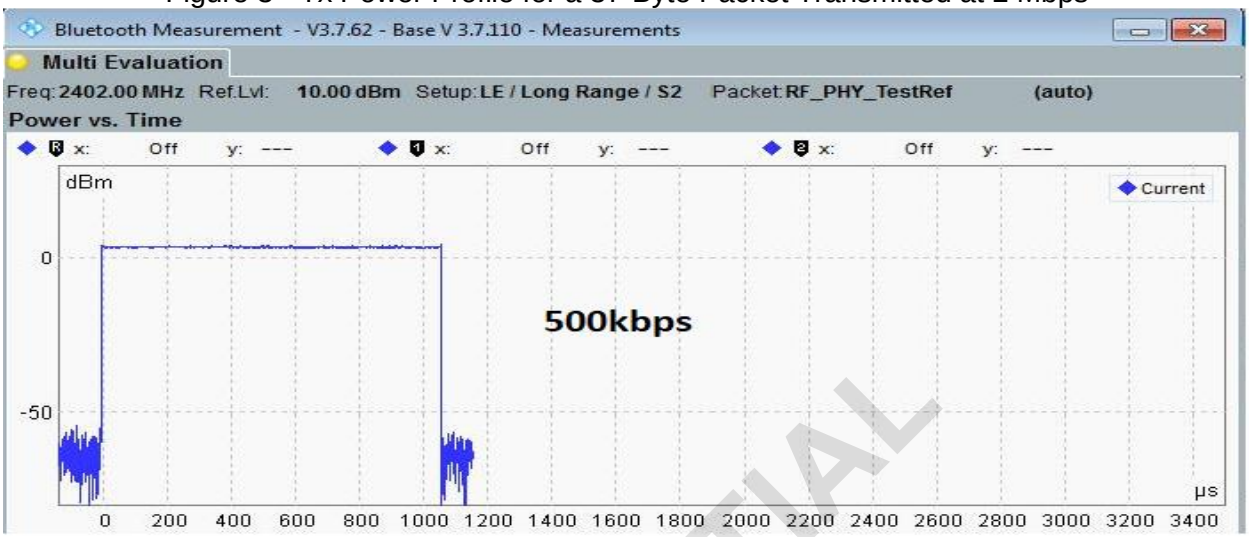


Figure 4 - Tx Power Profile for a 37-Byte Packet Transmitted at 500 kbps

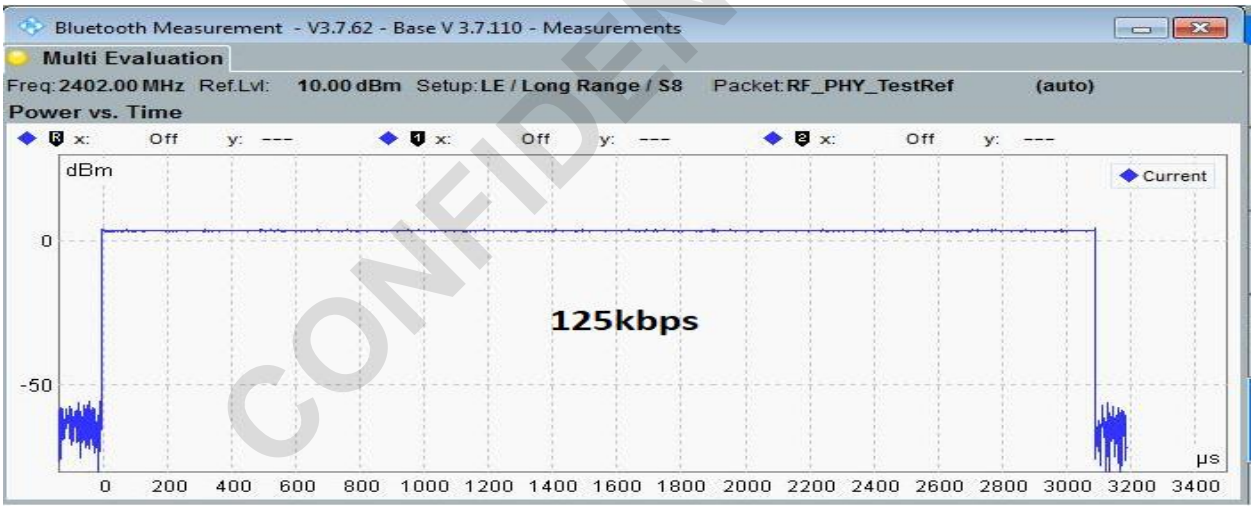


Figure 5 - Tx Power Profile for a 37-Byte Packet Transmitted at 125 kbps

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Features Supported by Atmosic's ATMxxx Series SoCs

Atmosic's ATMxxx series SoCs support the following optional features with the LE Coded PHY:

- Primary and/or secondary advertising using LE Coded PHY
- Scanning using LE Coded PHY
- Initiating a connection using LE Coded PHY
- PHY Update Procedure to change to LE Coded PHY after a connection is established
- Asymmetric connections where either transmit or receive is using LE Coded PHY
- GAP Manager (GAPM) supports multiple, concurrent non-connected activities: advertising, scanning, initiating, periodic synchronization activity
- GAP Controller (GAPC) supports multiple connections.

Enabling LE Coded PHY in Atmosic's ATMxxx Series SoCs

Atmosic's SDK release xx.xx (typically located at Customer Gateway under ATM3_2X/SW) provides detailed information on how to enable LE Coded PHY in ATMxxxx series devices. The documentation also provides examples of most-common use-cases such as beacons and Bluetooth LE advertising, either from OTP or external flash. A number of pre-configured beacon examples are provided in the makefile platform/atm2/ATM2xE3a/examples/OTP_beacon/reference_beacons.mk file. A desired reference beacon can be selected by setting the REF_BCN makefile variable when building nvds. For example, to enable 125 kbps and 500 kbps, do the following:
make REF_BCN:=LR_coded rebuild_otp_nvds
make REF_BCN:=LR500_coded rebuild_otp_nvds

The SDK already provides a variety of TDS source samples for common beacon contents and parameters. Moreover, those samples, together with the makefile targets pull_otp_nvds, rebuild_otp_nvds, and push_otp_nvds, are very useful in modifying NVDS data in a beacon device's OTP to change its behavior. The BLE_adv example is a flash-based version of the ROM OTP_beacon application that can access NVDS data in flash. It allows for rapid development and testing of OTP_beacon configuration parameters without needing to frequently replace devices with fully-burned OTP memories. See ../OTP_beacon/README for more information. It can also be used as a starting point for more complicated beacon applications such as those that are connectable and those that gather information from sensors and embed the data in advertisements.

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Frequently Asked Questions:

1. Do Atmosic's ATMxxxx series SoCs support LE long range ? How to enable long range mode?

   Yes, ATMxxxx series SOCs do support LE long range. For an example, please see the REF_BCN:=LR_coded config for BLE_adv and OTP_beacon. More info in OTP_beacon/reference_beacons.mk and OTP_beacon/tag_data/06-APP_BLE_ACT_CRT_CMD/ext_LR125_nc_ns_1s.tds.

2. Do Atmosic's ATMxxxx series SoCs support Advertising Extensions?

   Yes, Atmosic's ATMxxxx series SoCs do support Advertising Extensions. For an example, please see the REF_BCN:=2M_secondary config which uses an advertisement close to one thousand bytes long. OTP_beacon/tag_data/0b-APP_BLE_ADV_DATA/ext_huge_beacon_201.tds

3. Is there a solution to solve the bus conflict when running code in flash while writing to flash at the same time?

   Select uses of RAM for interrupt handlers are real-time performance optimizations. The Atmosic's QSPI bridge hardware and flash driver already support writing to and even erasing flash while executing in place from flash.

4. Where can I find more information about multiple advertising/scanning and multiple connections from a central device?

   Please see sections 4.7 and 5 in the SDK platform/atm2/ATM2xE3a/doc/ble/RW-BLE-GAP-IS.pdf document for GAPM host interface and GAPC controller interface.

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